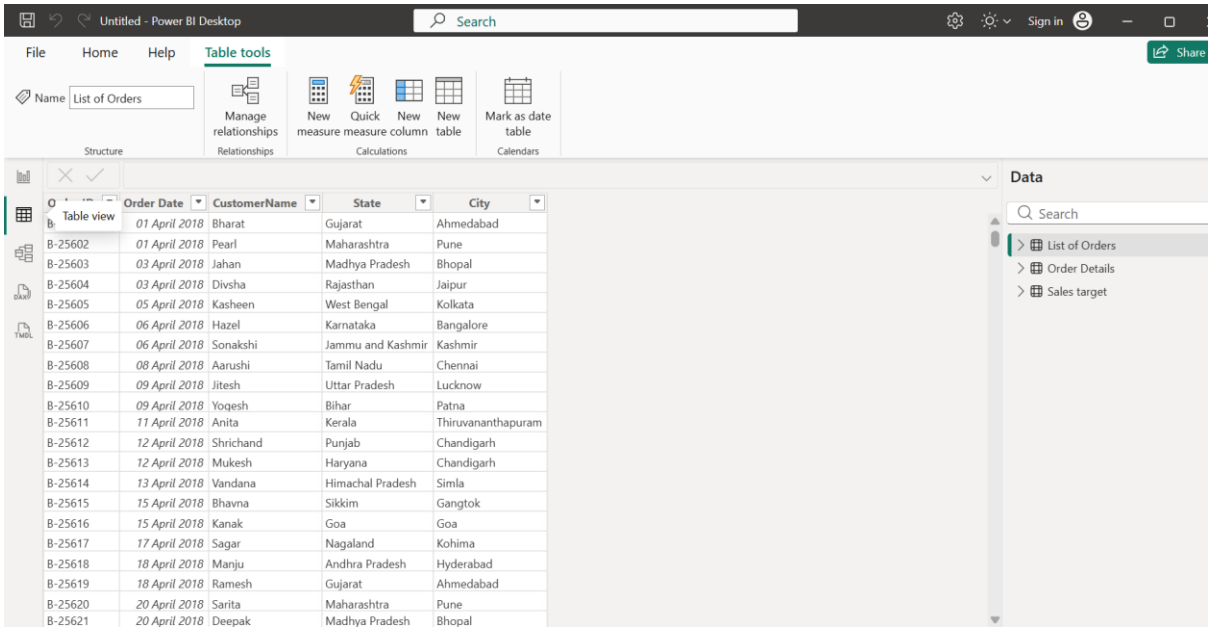
Module 2 – Assignment 1

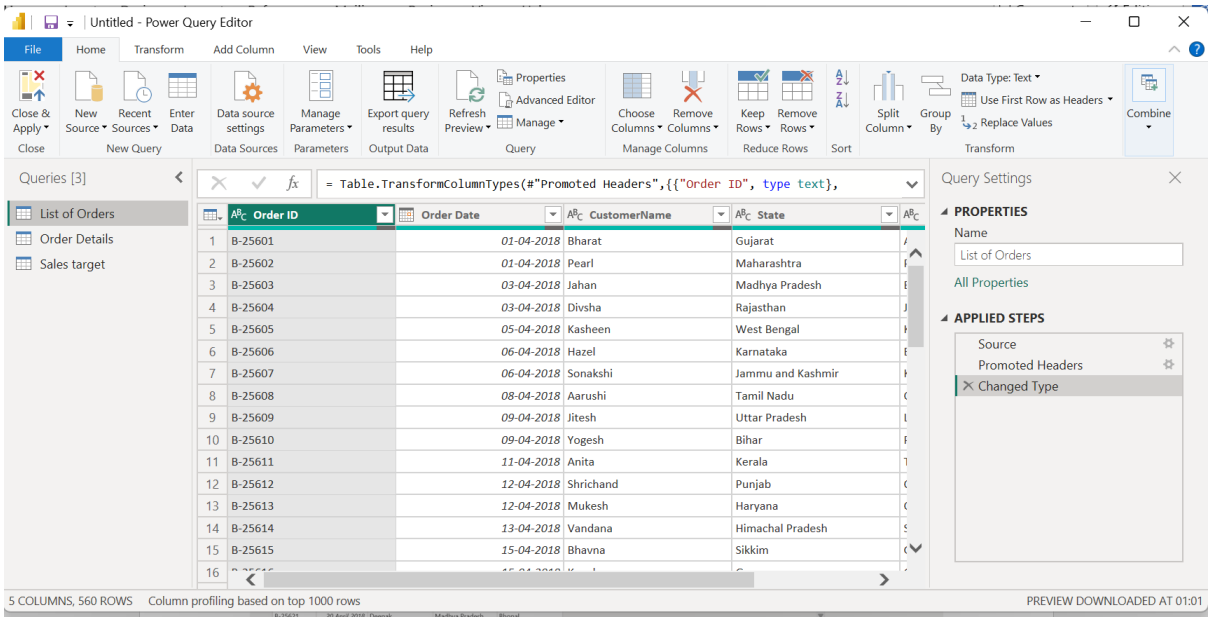
1. Data Import:

○ Import List of Orders.csv into Power BI.

○ Open the list of orders in the Power Query Editor using the Transform Data option.

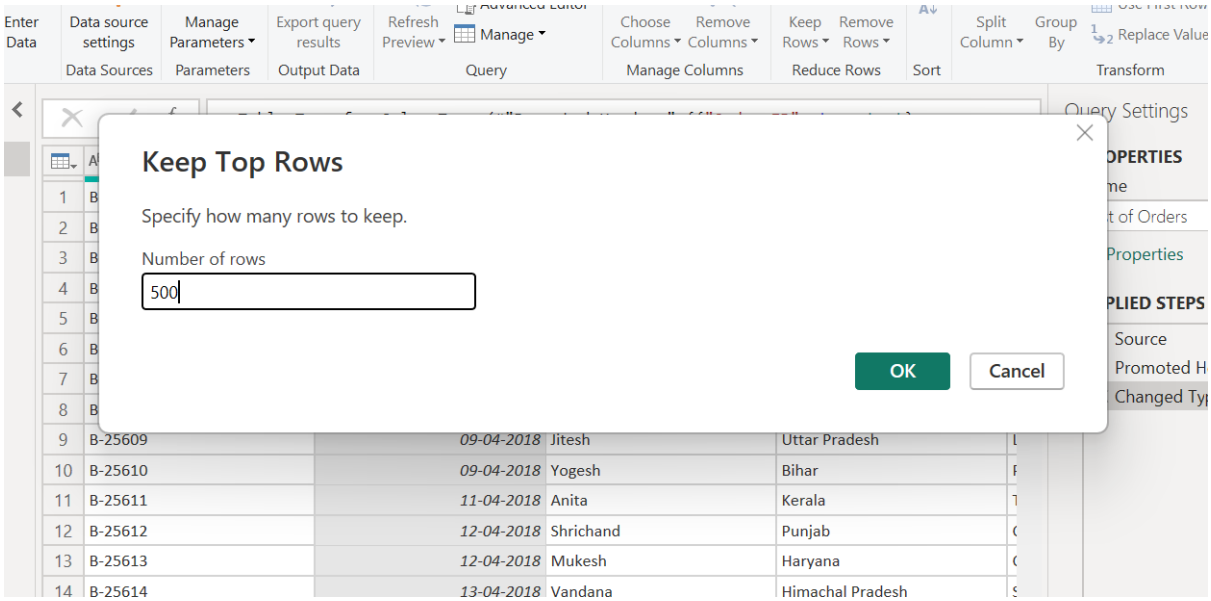
○ Import Order Details.csv and Sales Target.csv into Power Query Editor.



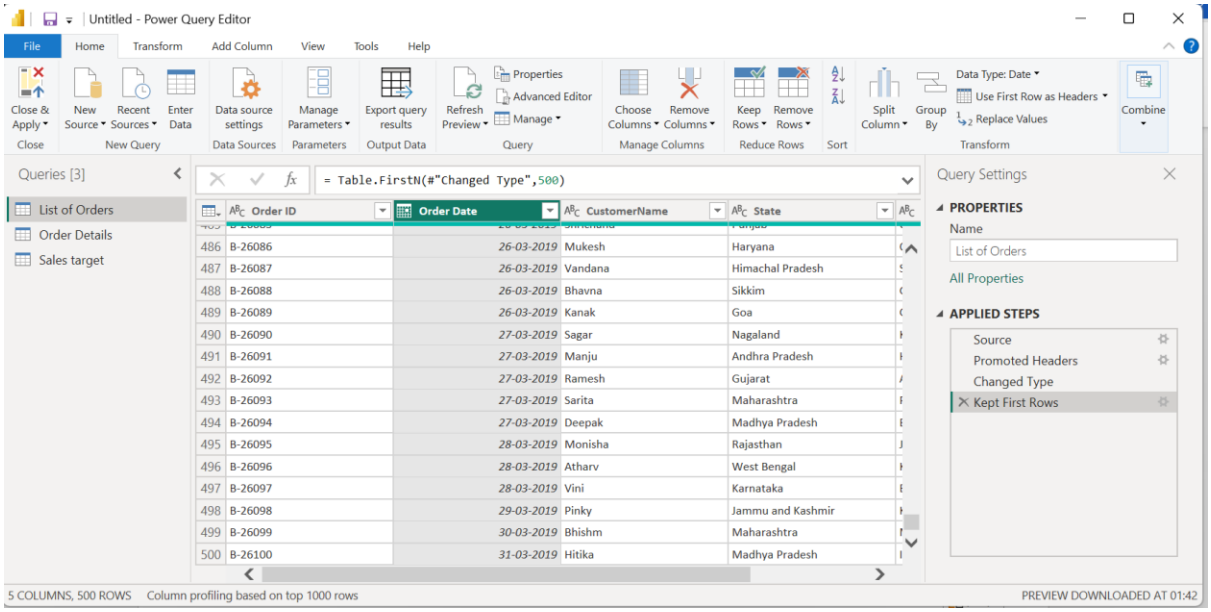


2. Row Limiting & Data Types:

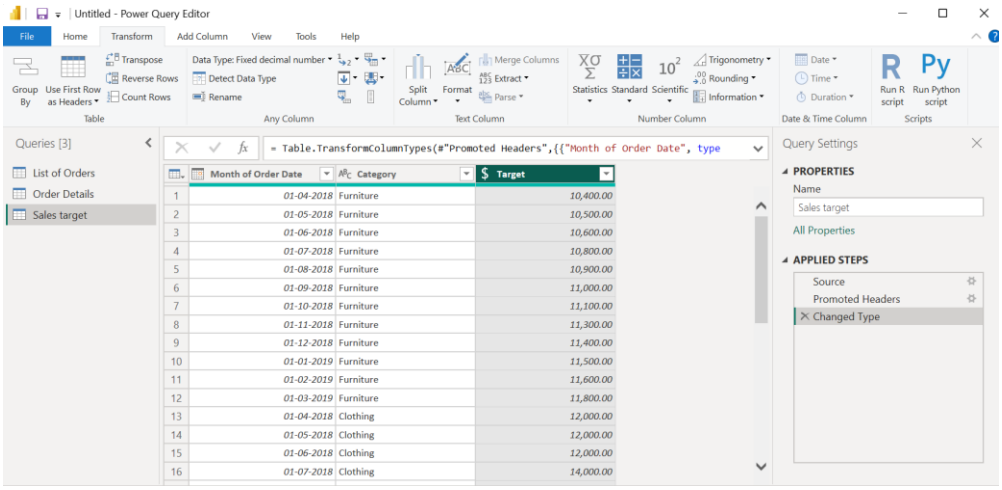
○ Restrict the List of Orders table to the first 500 rows.

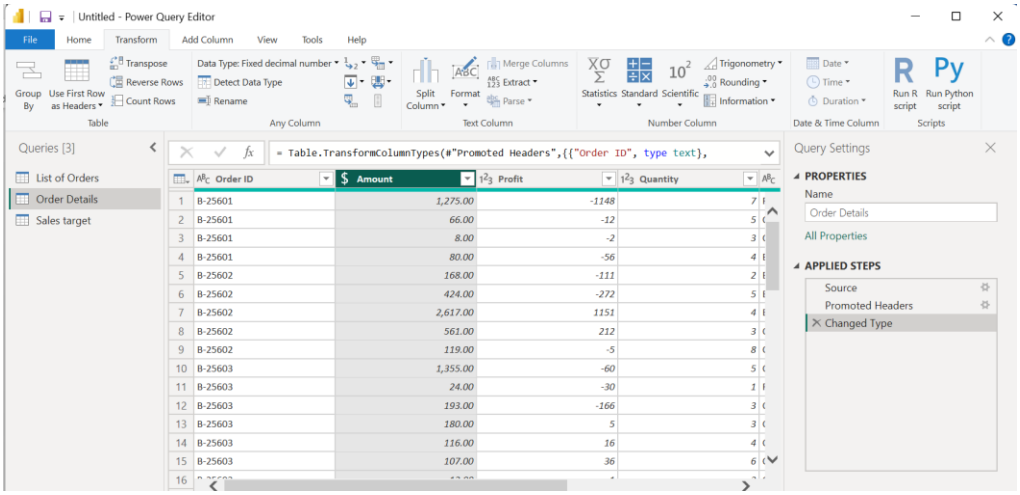


○ Set the Order Date column to the Date data type.



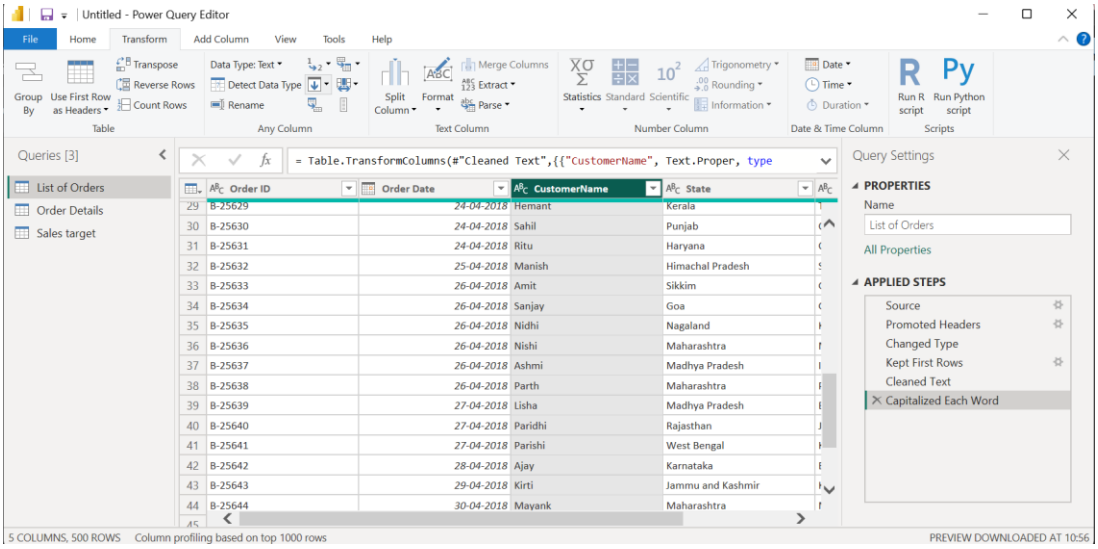
○ **Change the Amount and Target columns to fixed decimal numbers.**





3. Text Formatting:

○ **Format the CustomerName column to proper case to ensure consistent capitalization.**

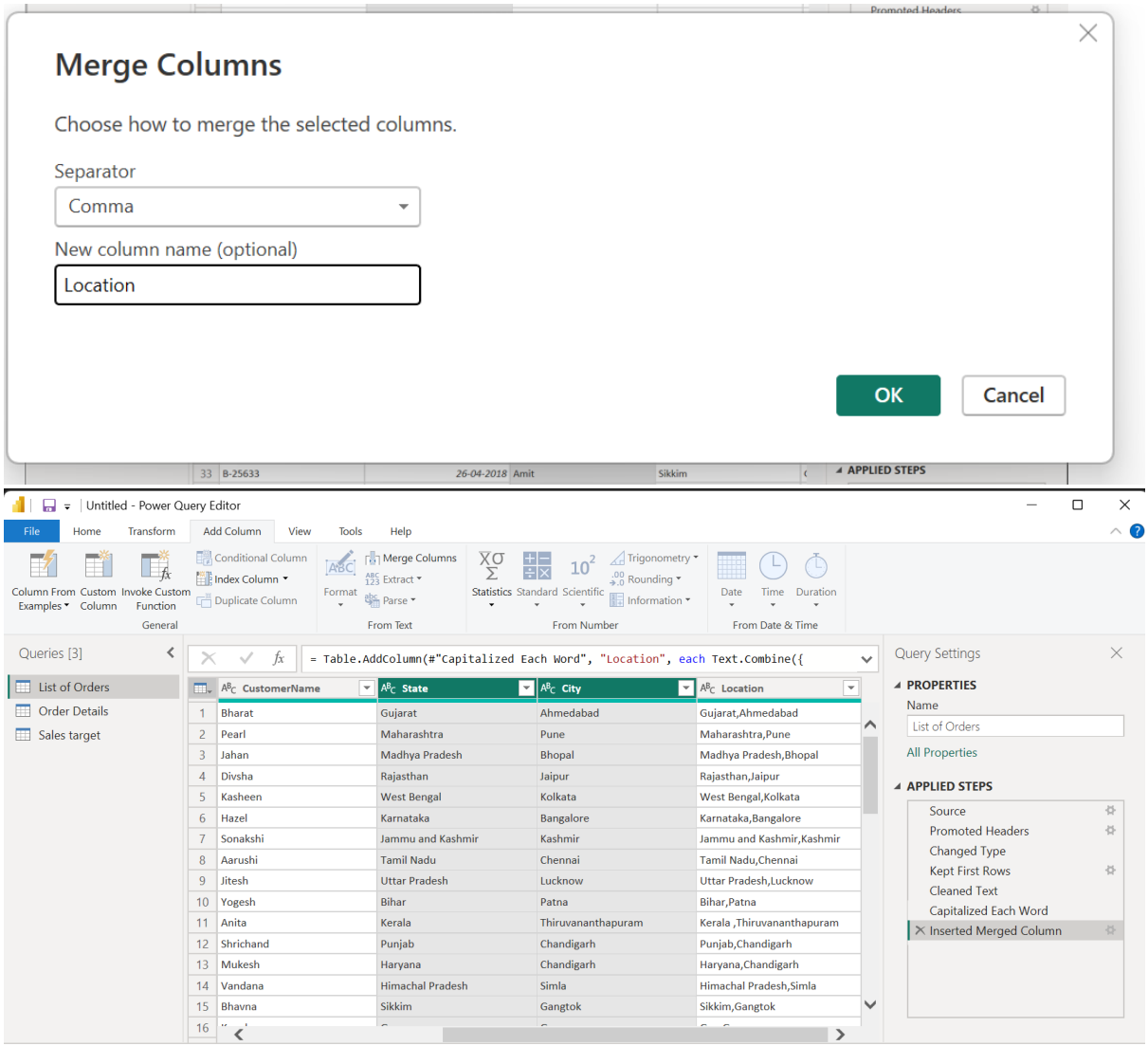


4.

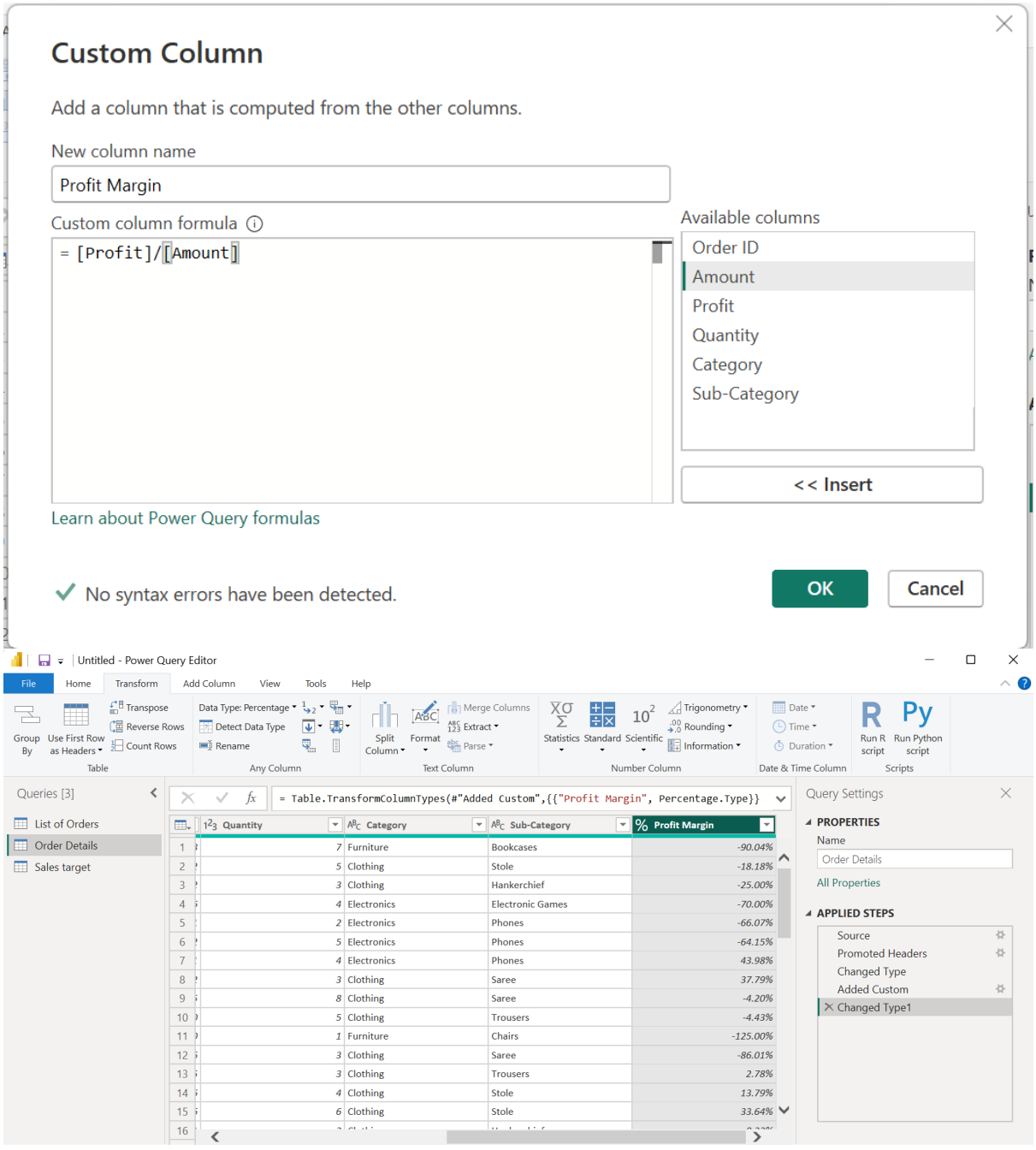
Column Creation:

○ Merge the City and State columns to create a new column named Location in the format:

City, State



○ **Create a custom column named Profit Margin calculated as: Profit / Amount (expressed as a percentage)**



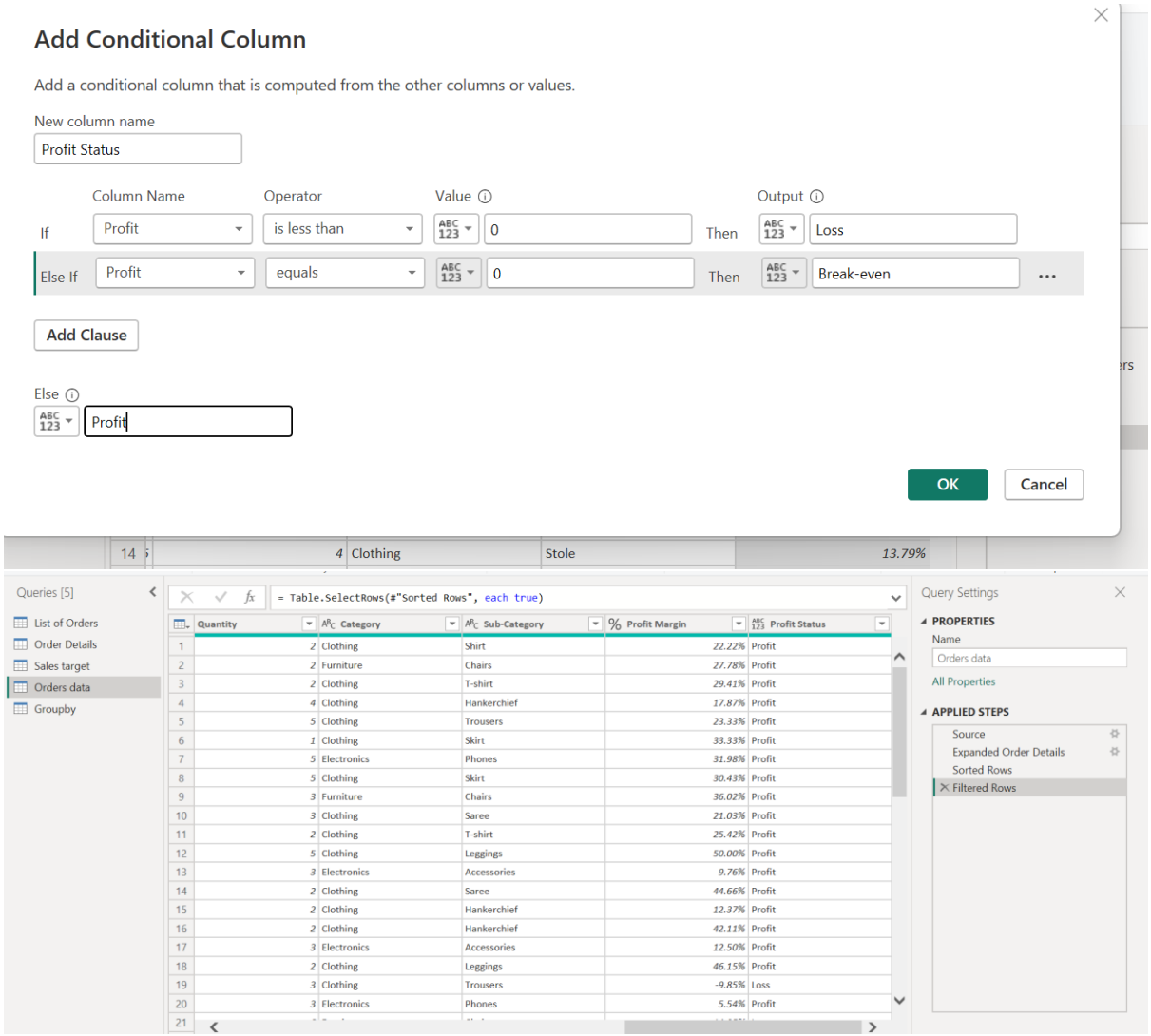
5. Conditional Column:

○ Add a conditional column named Profit Status based on:

■ Profit < 0 → Loss

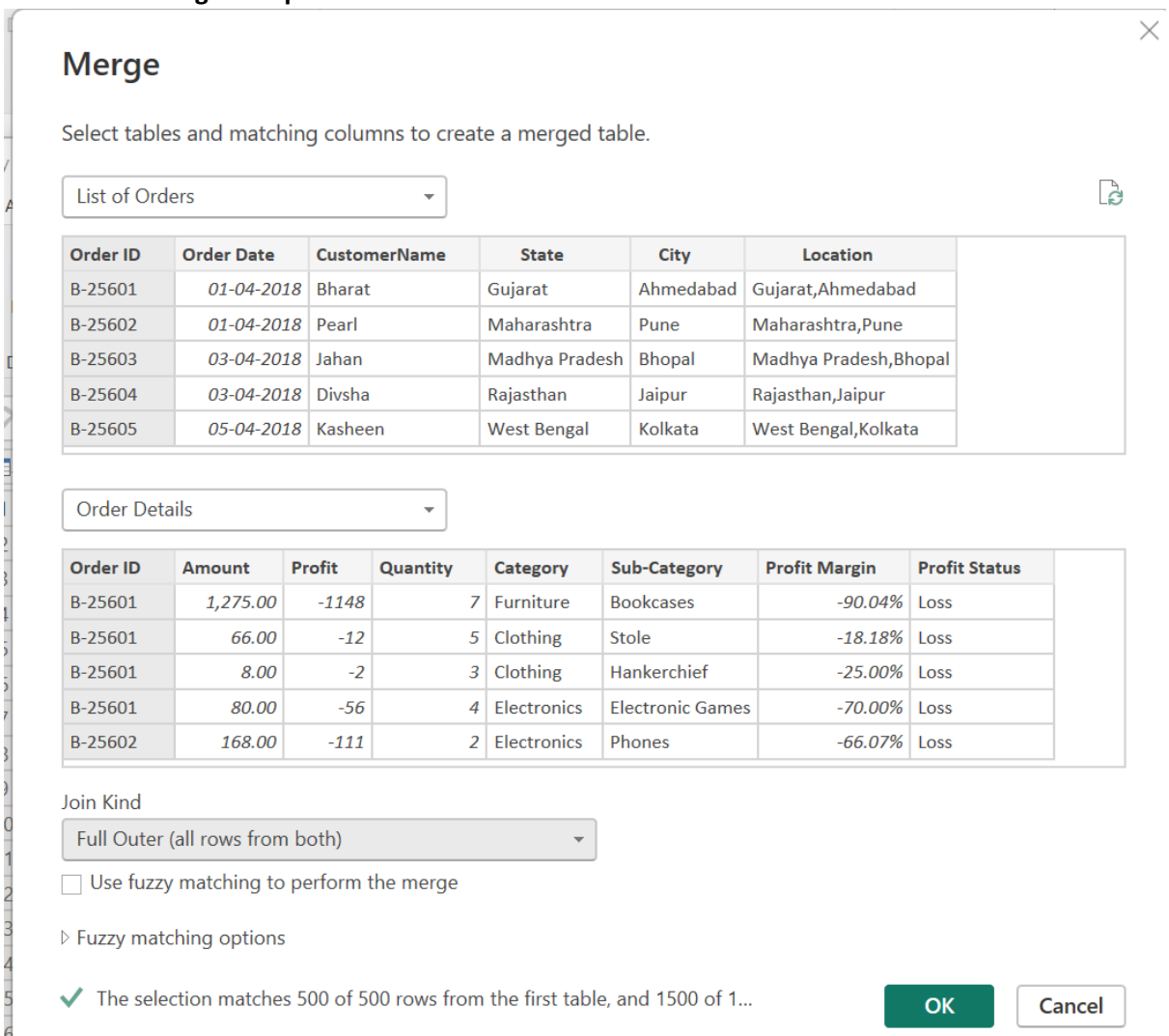
■ Profit = 0 → Break-Even

■ Profit > 0 → Profit



6. Merging Data (Joins)

- **Merge the List of Orders and Order Details tables using Order ID.**
- **Name the merged output table as Orders Data.**



Merge

Select tables and matching columns to create a merged table.

List of Orders

Order ID	Order Date	CustomerName	State	City	Location
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Gujarat,Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra	Pune	Maharashtra,Pune
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Madhya Pradesh,Bhopal
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur	Rajasthan,Jaipur
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata	West Bengal,Kolkata

Order Details

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status
B-25601	1,275.00	-1148	7	Furniture	Bookcases	-90.04%	Loss
B-25601	66.00	-12	5	Clothing	Stole	-18.18%	Loss
B-25601	8.00	-2	3	Clothing	Hankerchief	-25.00%	Loss
B-25601	80.00	-56	4	Electronics	Electronic Games	-70.00%	Loss
B-25602	168.00	-111	2	Electronics	Phones	-66.07%	Loss

Join Kind

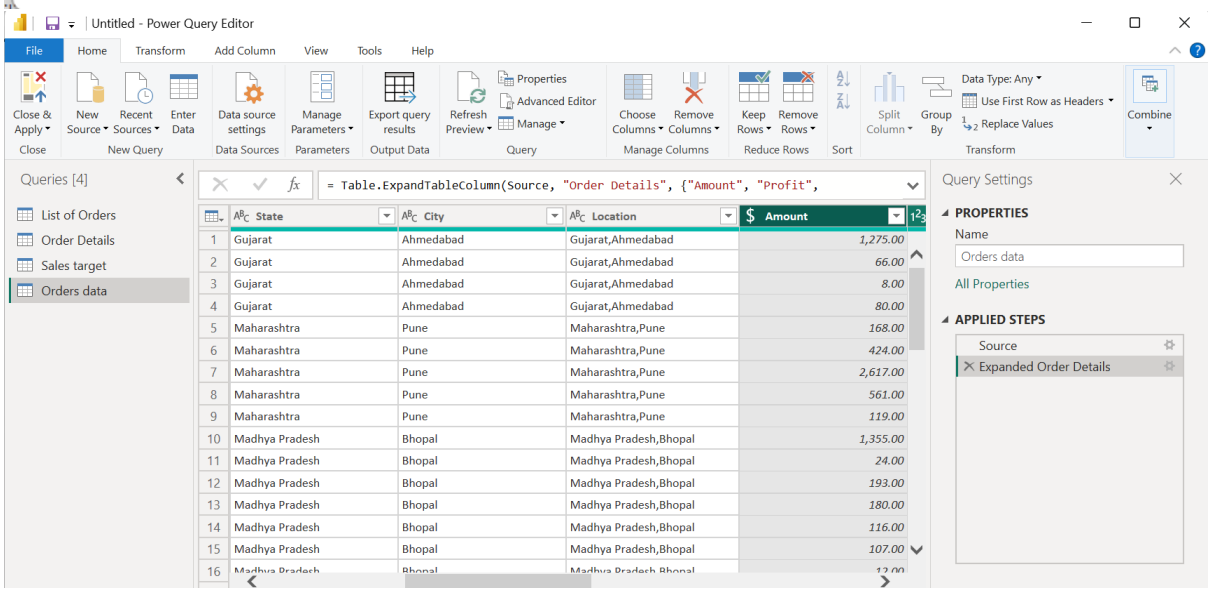
Full Outer (all rows from both)

Use fuzzy matching to perform the merge

Fuzzy matching options

The selection matches 500 of 500 rows from the first table, and 1500 of 1...

OK Cancel



Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply | New Source | Recent Sources | Enter Data | Data source settings | Manage Parameters | Export query results | Refresh Preview | Properties | Advanced Editor | Manage | Choose Columns | Remove Columns | Keep Rows | Remove Rows | Split Column | Group By | Data Type: Any | Use First Row as Headers | Replace Values | Combine

Close | New Query | Data Sources | Parameters | Output Data | Query | Manage Columns | Reduce Rows | Sort | Transform

Queries [4]

- List of Orders
- Order Details
- Sales target
- Orders data

= Table.ExpandTableColumn(Source, "Order Details", {"Amount", "Profit",

	State	City	Location	Amount
1	Gujarat	Ahmedabad	Gujarat,Ahmedabad	1,275.00
2	Gujarat	Ahmedabad	Gujarat,Ahmedabad	66.00
3	Gujarat	Ahmedabad	Gujarat,Ahmedabad	8.00
4	Gujarat	Ahmedabad	Gujarat,Ahmedabad	80.00
5	Maharashtra	Pune	Maharashtra,Pune	168.00
6	Maharashtra	Pune	Maharashtra,Pune	424.00
7	Maharashtra	Pune	Maharashtra,Pune	2,617.00
8	Maharashtra	Pune	Maharashtra,Pune	561.00
9	Maharashtra	Pune	Maharashtra,Pune	119.00
10	Madhya Pradesh	Bhopal	Madhya Pradesh,Bhopal	1,355.00
11	Madhya Pradesh	Bhopal	Madhya Pradesh,Bhopal	24.00
12	Madhya Pradesh	Bhopal	Madhya Pradesh,Bhopal	193.00
13	Madhya Pradesh	Bhopal	Madhya Pradesh,Bhopal	180.00
14	Madhya Pradesh	Bhopal	Madhya Pradesh,Bhopal	116.00
15	Madhya Pradesh	Bhopal	Madhya Pradesh,Bhopal	107.00
16	Madhya Pradesh	Bhopal	Madhya Pradesh,Bhopal	12.00

Query Settings

PROPERTIES

Name: Orders data

All Properties

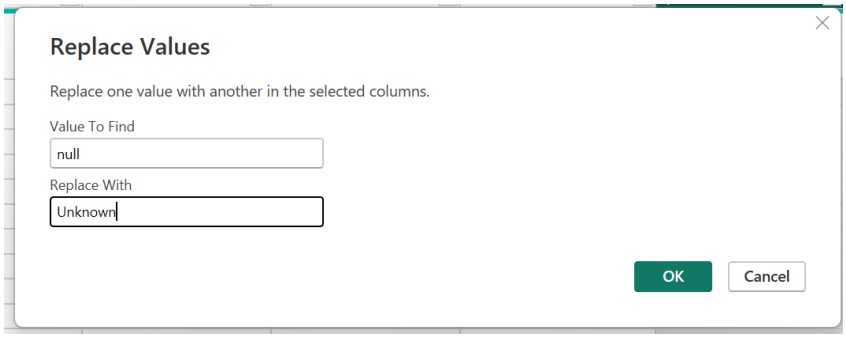
APPLIED STEPS

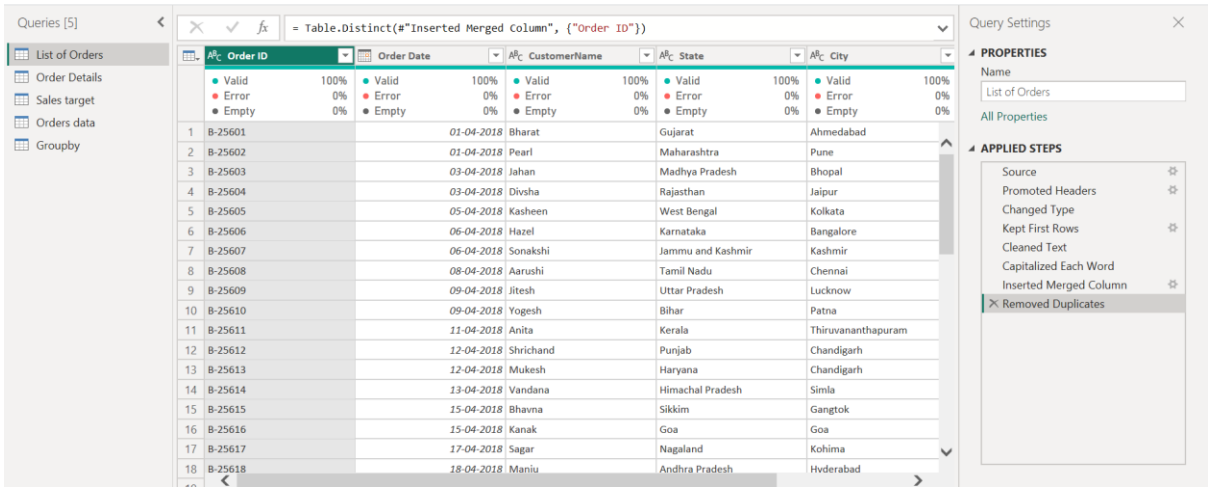
- Source
- Expanded Order Details

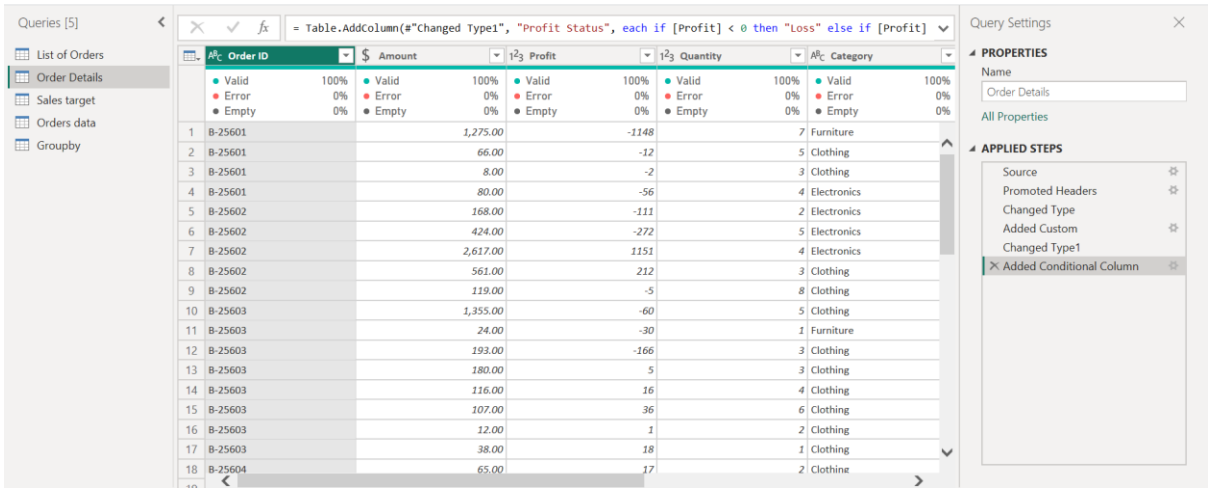
Handling Missing Data & Duplicate Data

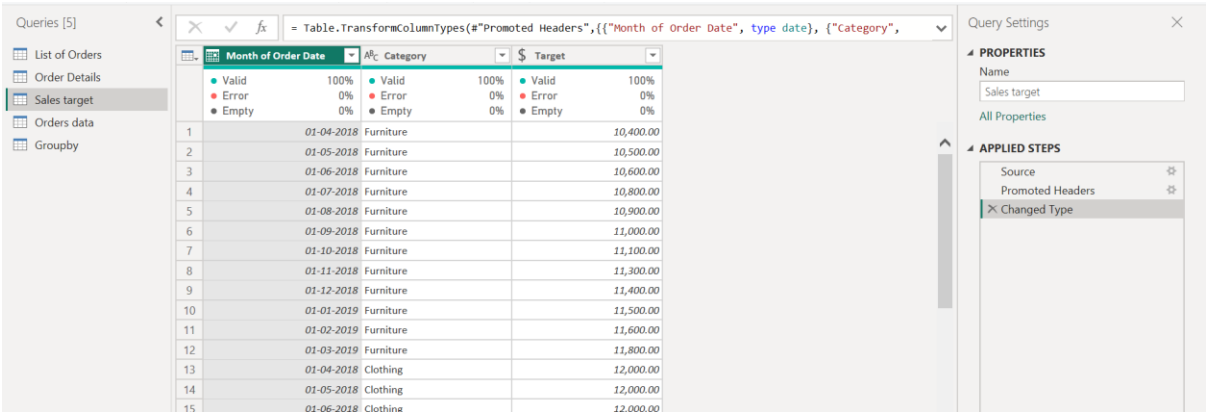
- **Identify missing values across all tables.**
- **Define and apply appropriate strategies such as:**
  - **Replacing missing values**
  - **Removing records**
  - **Retaining nulls with justification**

There is no empty in the tables . This was identified by using the column quality. If there is any null column we can replace it with the suitable data.



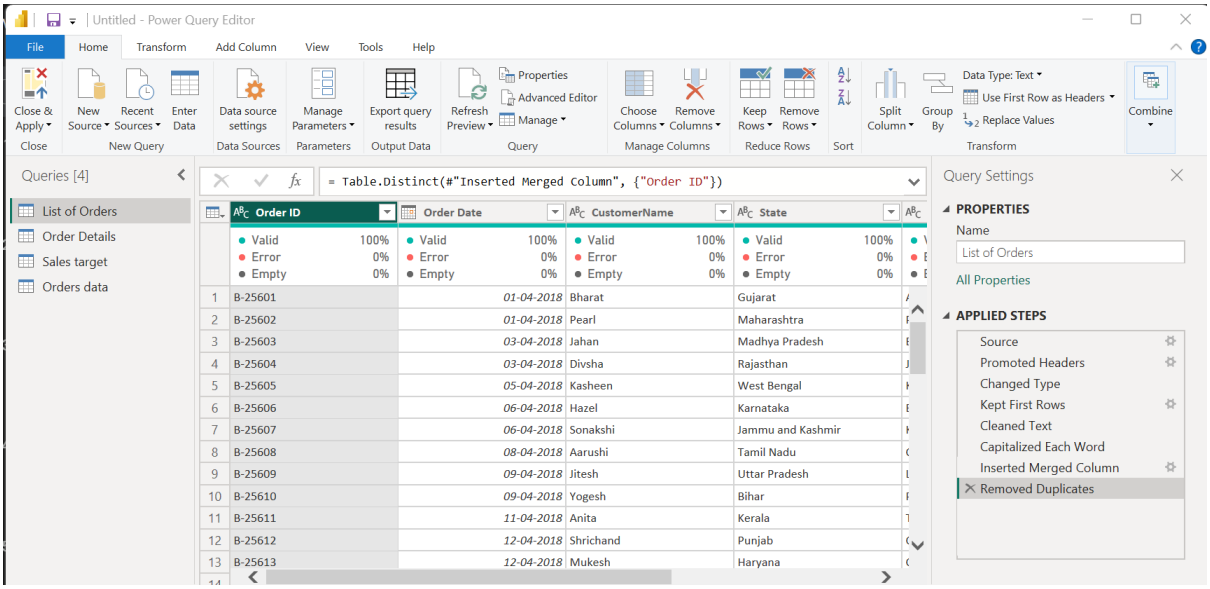






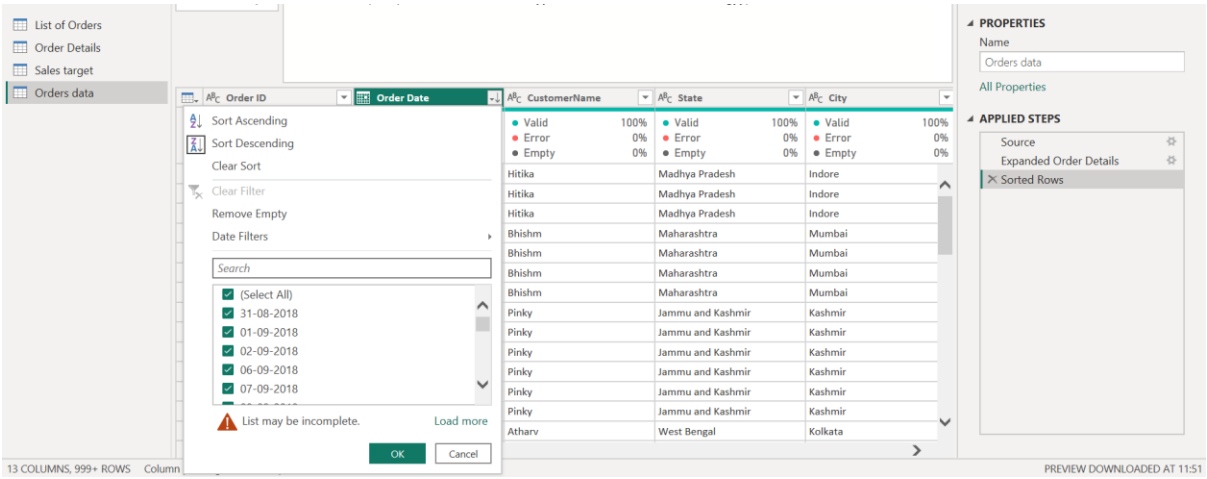
● Identify duplicate rows and define a suitable handling strategy based on business logic.

Based on the order ID the duplicates are removed



8. Sorting and Filtering Data

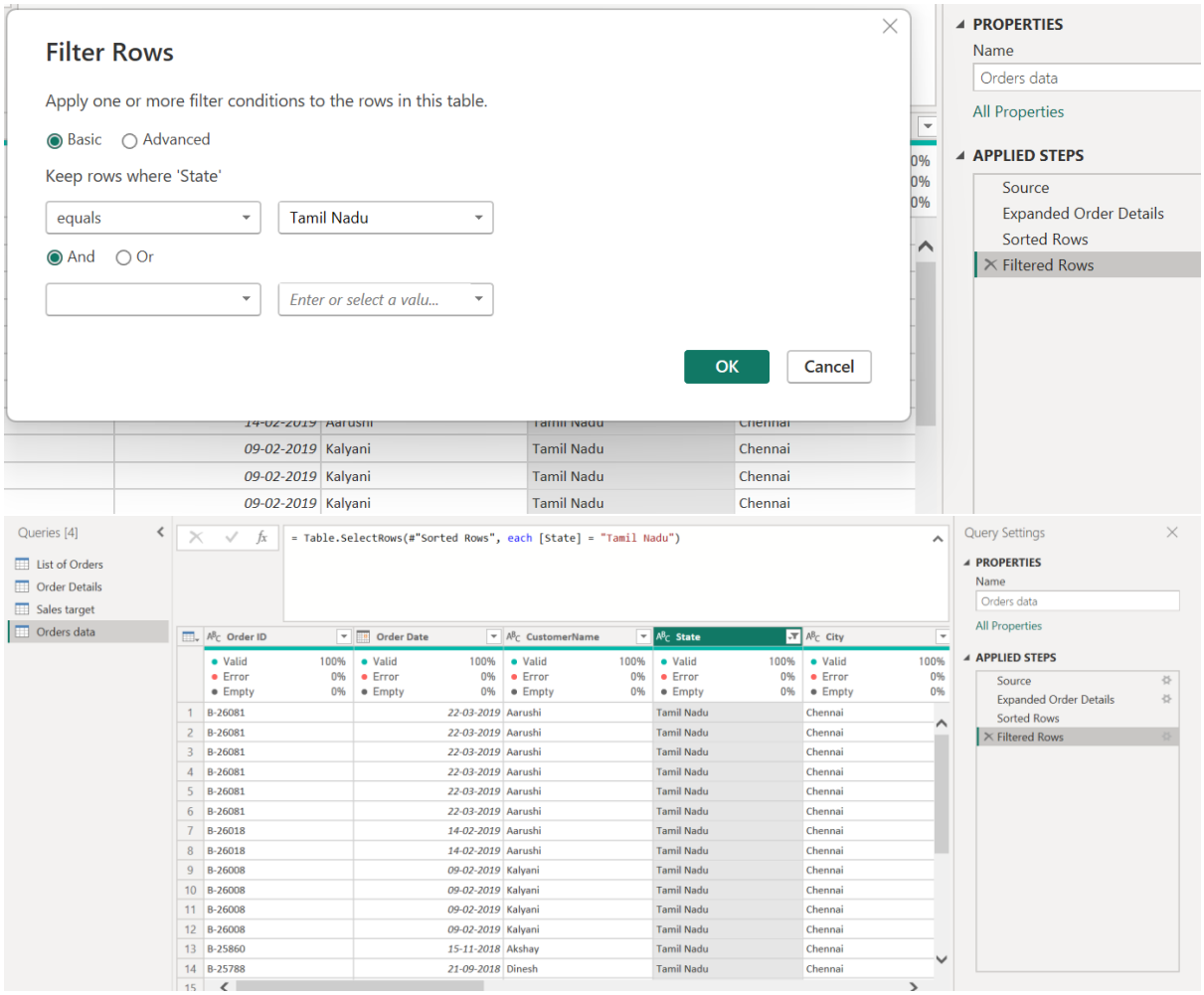
● Sort records in the Orders Data table by Order Date in descending order.



● Apply filters to analyze orders from a specific state (e.g., Tamil Nadu).

● Apply filters to analyze orders from a specific state (e.g., Tamil Nadu).

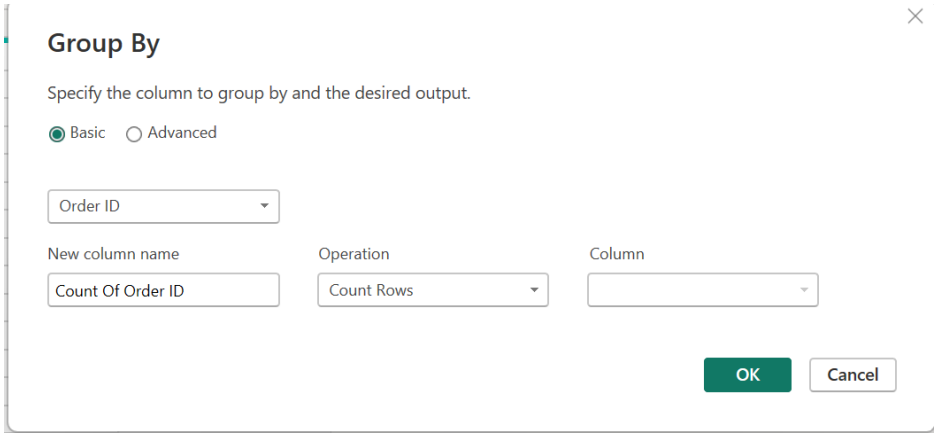
- **Apply filters to analyze orders from a specific state (e.g., Tamil Nadu).**

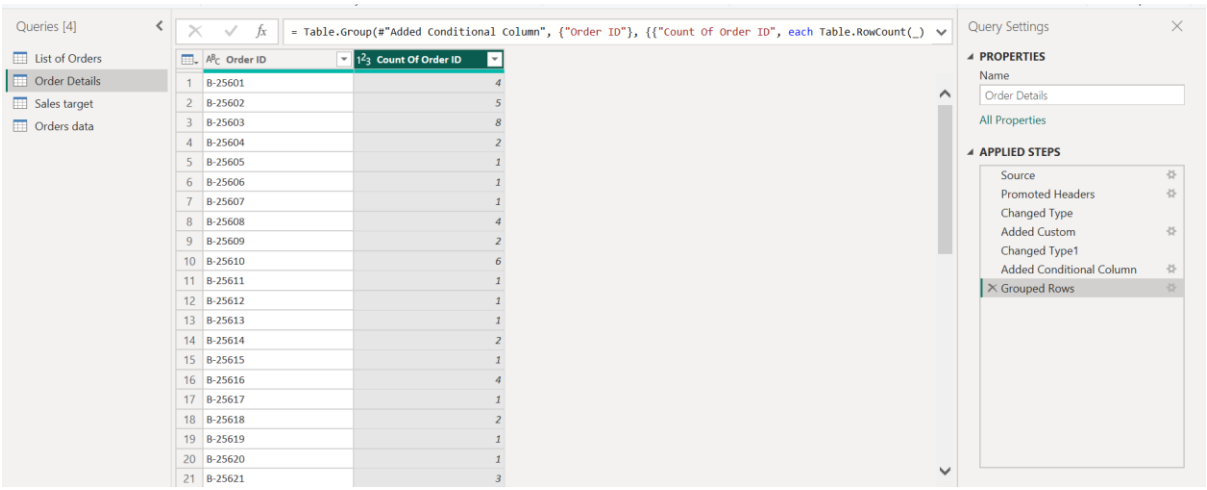


9. Grouping and Aggregating Data

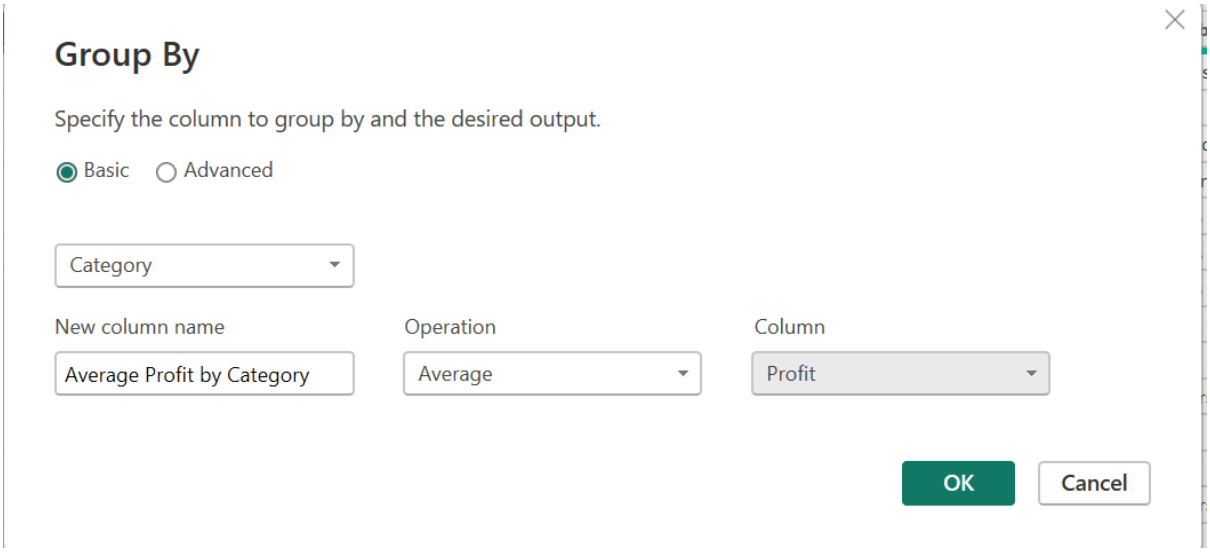
- **Duplicate the Order Details table and perform the following:**

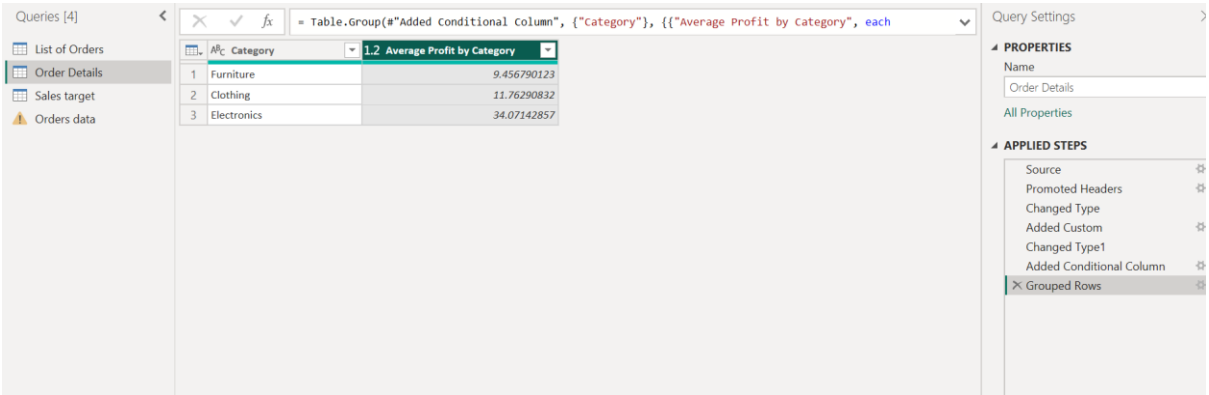
- **Count of Order ID**



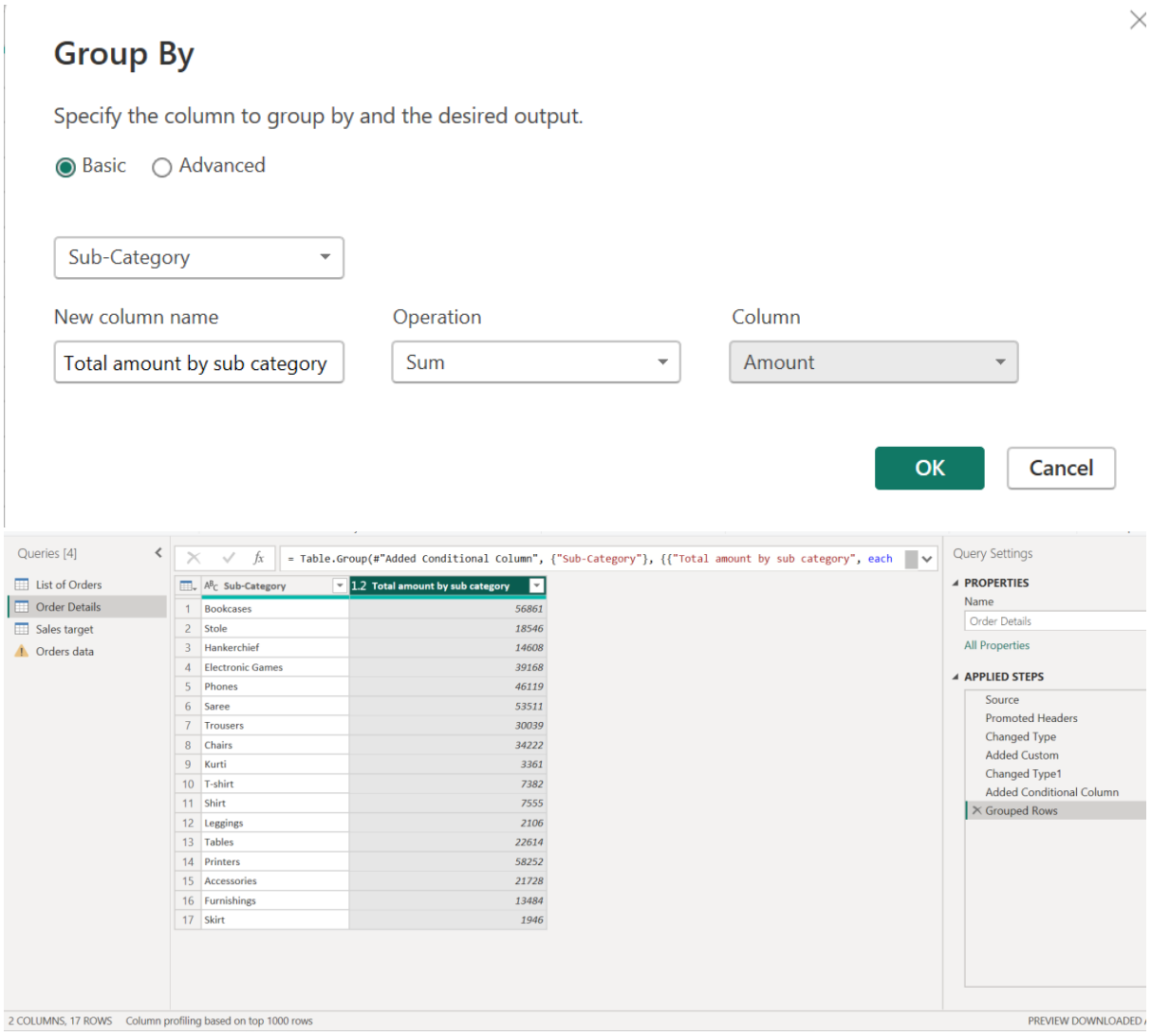


- **Average Profit by Category**

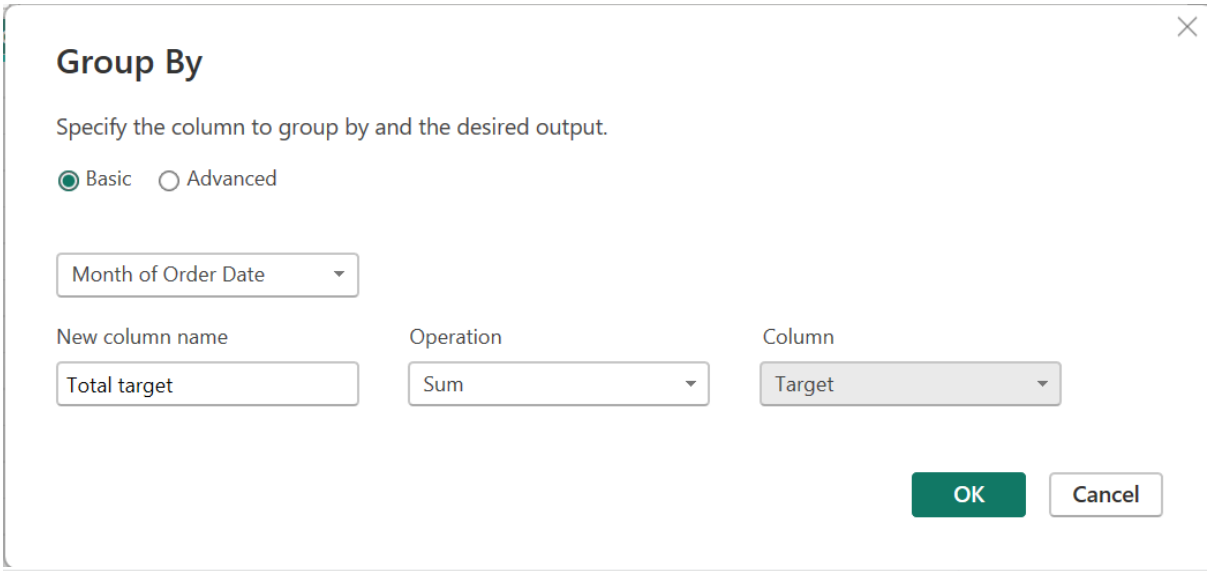


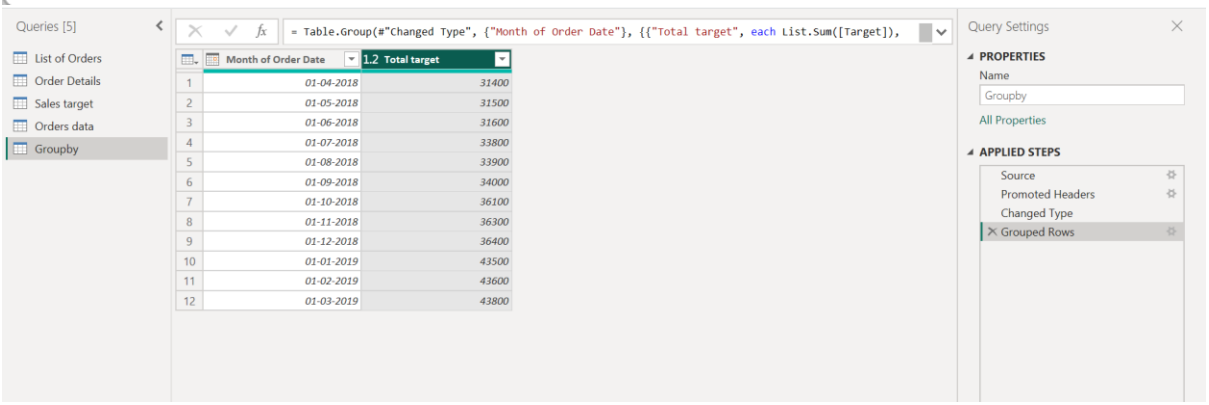


- **Total Amount by Sub-Category**



● Duplicate the Sales Target table and aggregate the total Target by Month of order date.

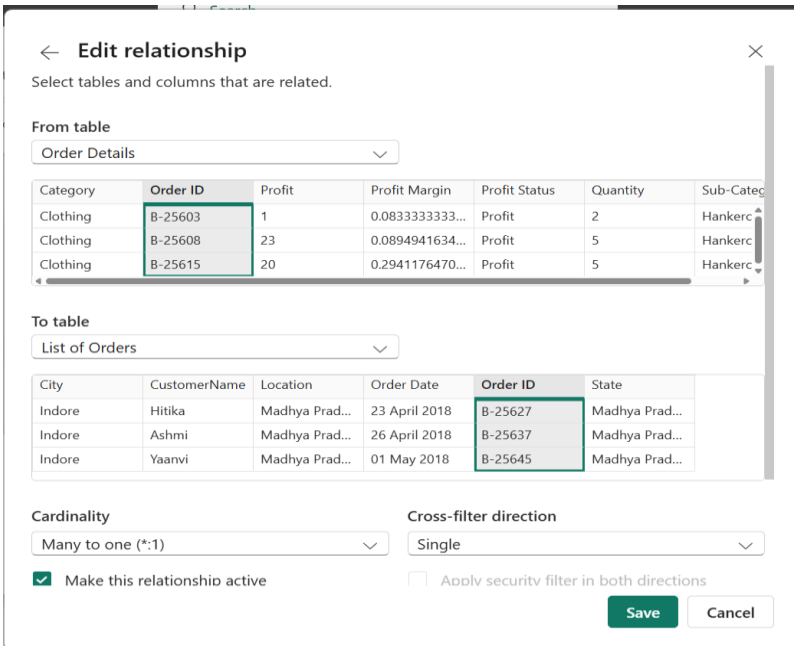




10. Data Modeling:

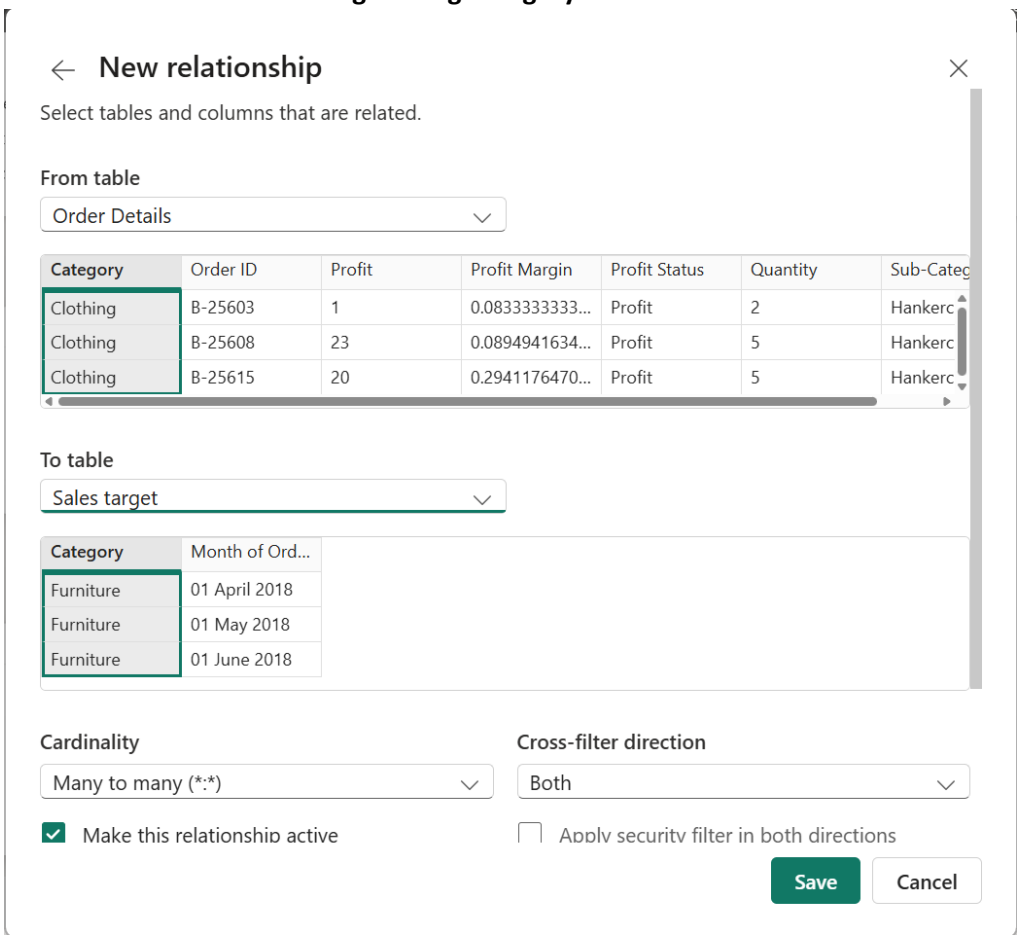
● Establish a relationship between:

○ List of Orders and Order Details using Order ID

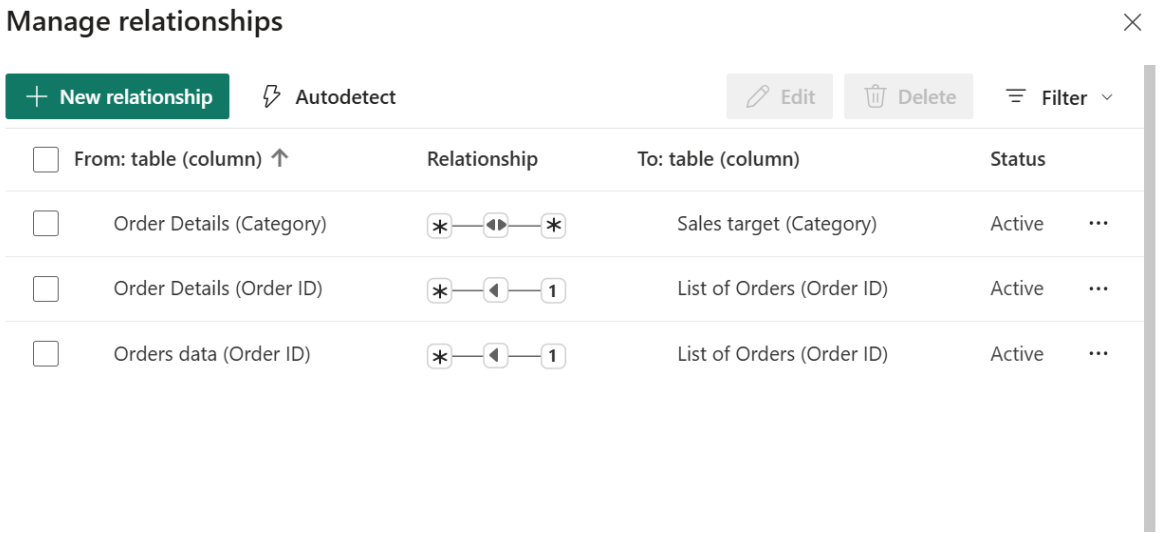


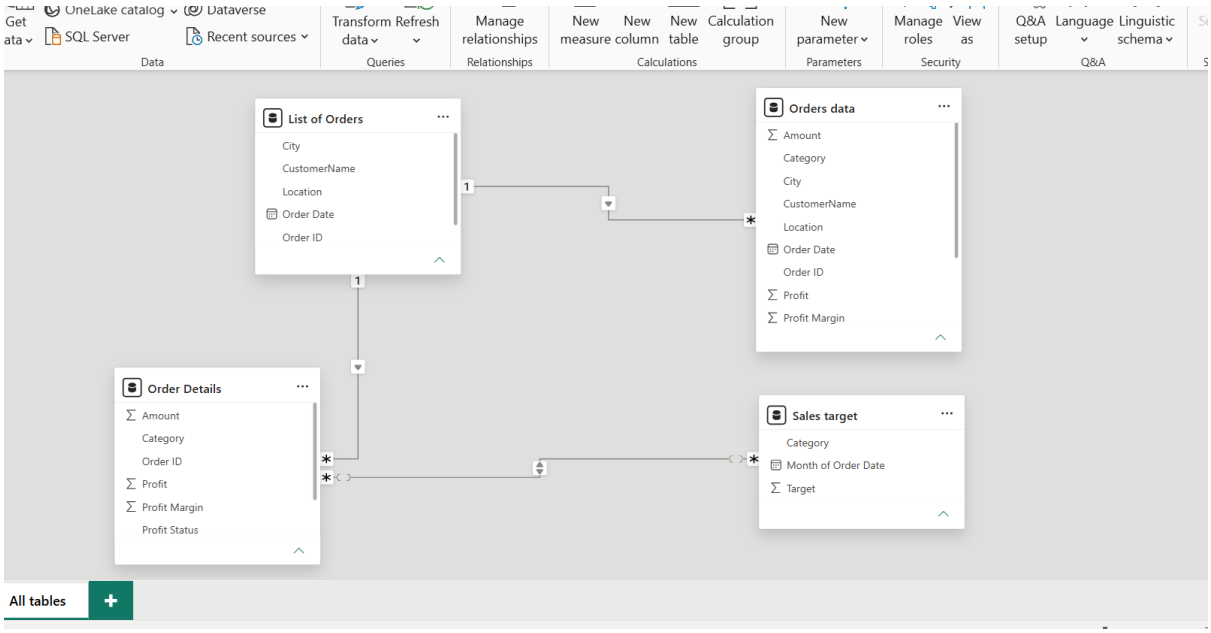
- **Establish a relationship between:**

  - **Order Details and Sales Target using Category**



2. Use Manage Relationships to ensure relationships are active and correctly defined.




Get data
OneLake catalog
Dataverse
SQL Server
Recent sources
Data
Transform data
Refresh
Queries
Manage relationships
Relationships
New measure
New column
New table
Calculation group
Calculations
New parameter
Parameters
Manage roles
View as
Security
Q&A setup
Language
Linguistic schema
Q&A
List of Orders
City
CustomerName
Location
Order Date
Order ID
Orders data
Amount
Category
City
CustomerName
Location
Order Date
Order ID
Profit
Profit Margin
Order Details
Amount
Category
Order ID
Profit
Profit Margin
Profit Status
Sales target
Category
Month of Order Date
Target
1
*
All tables